1. → React is a Javascript library used to build user interfaces UI, especially website.
→ Create reusable parts of a website called Components
→ Component-based architecture; Dividing a website into small reusable parts like Navbar, Header, Footer.
→ Declarative UI: We tell React what we want to display, and React updates the UI.

2. → JSX stands for JavaScript XML. It allow us to write HTML-code inside JavaScript.
→ Browsers Cannot directly understand JSX, so tool like Babel or Vite transform JSX into JavaScript.
→ This process is JSX transformation, we can write JavaScript expression inside JSX using {}.
→ JSX use className="" instead of class HTML's, all tag Properly closed. Const name = "Teju"; <h1> Hello {name} </h1>

3. → Real DOM - the actual structure of a webpages displayed in the browser
→ Virtual DOM - is lightweight JavaScript representation of UI
→ React Compares the perviors and new Virtual DOM. the Process is reconciliation
→ React identifies what changed and update only the necessary parts of the Real DOM.
→ helps avoid unnecessary DOM updates.

4. → Rendering → means React Creates the UI based on the Component's Code
→ A Component re-renders when it state changes or when it receives new props.
→ A parent Components re-rendering can also cause its

child components to re-render.
→ React check the new UI and update the actual DOM only when necessary.
→ Re-rendering does not always mean the entire webpage change.

5. → props are data passed from a parent component to child components
→ state is data managed inside a component that can change over time.
→ React follows one-way data flow, meaning data generally moves from parents to child.
→ lifting state up - means moving shared state to a common parent component.
→ State not directly modified because React may not detect change. We should use the State setter function.
→ useState() is React Hook used to store and update data inside a component.
→ It return two value : The current state and a function to update it.
→ state is as snapshot, meaning each render has its own state value.
→ React may combine multiple State update into one render. This called batching.

→ If the next state depends on the previous state, use a functional update like setCount(prev => prev+1)

const [count, setCount] = useState(0);

7. → useEffect() is a React Hook used to handle side effect like API calls, timers and event listeners.

→ The dependencies array control when the effect run: no arrays means after every render, [] means after the initial mount, and [value] means when that value changes.

→ A clean up function removes timers or event listeners when they are no longer needed.

→ A state colusre happen when an effect uses an old value from an earlier render.

→ In development strict mode, React runs an extra setup and clean up cycle to help identify problems.

~~what~~ → Controlled components are form elements whose value are managed by React state.

→ Uncontrolled components store their values inside the DOM instead of React state.

→ Controlled components are useful for form validation and tracking user input.

→ useRef() help access an uncontrolled input's value directly.

→ Controlled inputs are useful for most forms, while uncontrolled inputs are useful for simple forms.

9.

→ React uses map() to display multiple items from an array

→ Each list item need a unique and stable key to help React identity it.

→ Using array indexes as key can cause problem when list are added, removed or reordered.

→ Conditional rendering means displaying content based on a condition using if, ternary (?:) or &&.

→ In JSX, string and numbers are displayed but true, false, null and undefined are not displayed.

Example:
```
{items.map(item => (
<p key={item.id}>{item.name}</p>
))}
```

→ React Router is a library used to navigate between different pages or components in a React application.

→ BrowserRouter enable browser-based routing, while Routes groups the route definitions.

→ Route define a path and its corresponding Components.

→ Link and NavLink navigate between pages with a full page reload. NavLink can also identify the active Link

→ Nested routes allow routes inside other routes, and outlet display the matching child route.

11. → use Navigate() is used to navigate from one Page to another using JavaScript.

→ replace: true replace the current history entry.

→ use Params() gets dynamic values from the URL Path, such as /user/10.

→ use Search Params() get or updates query parameters such as ?name=Teju.

→ Navigation state allow us to pass additional data from one page to another without putting it in the URL.